Homework Set # 9

S 7.4.2 $\quad |n\rangle = \dfrac{1}{\sqrt{n!}}\, a^{\dagger n}\, |0\rangle$

$$a = \frac{1}{\sqrt{2}}\left(\frac{1}{\Delta} X + \frac{i\Delta}{\hbar} P\right) \qquad \Delta = \left(\frac{\hbar}{m\omega}\right)^{1/2}$$

$$[a, a^\dagger] = 1$$

$$\langle n|X|n\rangle = \frac{1}{n!}\frac{\Delta}{\sqrt{2}} \langle 0| a^n (a + a^\dagger) a^{\dagger n} |0\rangle$$

$$= 0$$

since $\langle 0| a^n a^{\dagger n'} |0\rangle = 0$ if $n \neq n'$

$$\langle n|P|n\rangle = \frac{1}{n!}\frac{i\hbar}{\sqrt{2}\,\Delta} \langle 0| a^n (a^\dagger - a) a^{\dagger n} |0\rangle$$

$= 0$ for the same reason

$$\langle n|X^2|n\rangle = \frac{1}{n!}\frac{\Delta^2}{2} \langle 0| a^n (a + a^\dagger)^2 a^{\dagger n} |0\rangle$$

$$= \frac{1}{n!}\frac{\Delta^2}{2} \langle 0| a^n (a a^\dagger + a^\dagger a) a^{\dagger n} |0\rangle$$

$$\langle n|x^2|n\rangle = \frac{1}{n!}\frac{\Delta^2}{2}\langle 0|\, a^n (-1+2aa^\dagger)\, a^{\dagger n}|0\rangle$$

$$= \frac{\Delta^2}{2}\frac{1}{n!}\left(-n! + 2(n+1)!\right) = \frac{\Delta^2}{2}\left(-1+2(n+1)\right)$$

$$= \Delta^2 \left(n+\frac{1}{2}\right) = \frac{\hbar}{m\omega}\left(n+\frac{1}{2}\right)$$

$$\langle n|P^2|n\rangle = \frac{-1}{n!}\frac{\hbar^2}{2\Delta^2}\langle 0|\, a^n (a^\dagger - a)^2\, a^{\dagger n}|0\rangle$$

$$= \frac{+1}{n!}\frac{\hbar^2}{2\Delta^2}\langle 0|\, a^n (a^\dagger a + aa^\dagger)\, a^{\dagger n}|0\rangle$$

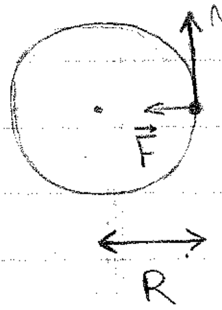
$$= \frac{\hbar^2}{\Delta^2}\left(n+\frac{1}{2}\right) = \hbar m\omega\left(n+\frac{1}{2}\right)$$

$$\Delta x\, \Delta p = \left[\langle n|x^2|n\rangle\langle n|P^2|n\rangle\right]^{1/2}$$

$$= \hbar\left(n+\frac{1}{2}\right)$$

7.4.3.

$$V(r) = a r^k$$



$$\frac{m v^2}{r} = F = \frac{dV}{dr} = a k r^{k-1}$$

$$\bar{T} = \overline{\frac{m}{2} v^2} = \frac{1}{2} a k r^k$$

$$= \frac{k}{2} \bar{V}$$

From S7.4.2

$$\langle T \rangle = \frac{1}{2m} \langle n | P^2 | n \rangle = \frac{\hbar\omega}{2}\left(n + \frac{1}{2}\right)$$

$$\langle V \rangle = \frac{m\omega^2}{2} \langle n | X^2 | n \rangle = \frac{\hbar\omega}{2}\left(n + \frac{1}{2}\right)$$

$$= \langle T \rangle$$

7.4.4

$$\langle n|x^4|n\rangle = \langle n|\frac{\Delta^4}{4}(a+a^\dagger)^4|n\rangle$$

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \langle n|\left[aa a^\dagger a^\dagger + a a^\dagger a a^\dagger + a^\dagger a a a^\dagger + a^\dagger a a^\dagger a + a^\dagger a^\dagger a a + a a^\dagger a^\dagger a\right]|n\rangle$$

(the other terms do not have equal numbers of a's and $a^\dagger$'s)

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \langle n|\left[6\, a a a^\dagger a^\dagger - (1+2+3+4+2)\, a a^\dagger + (1+2)\right]|n\rangle$$

since $a^\dagger a = a a^\dagger - 1$

$$a a^\dagger a a^\dagger = a a a^\dagger a^\dagger - a a^\dagger$$

$$a^\dagger a a a^\dagger = a a a^\dagger a^\dagger - 2 a a^\dagger$$

$$a^\dagger a a^\dagger a = a a a^\dagger a^\dagger - a^\dagger a - 2 a a^\dagger$$

$$= a a a^\dagger a^\dagger - 3 a a^\dagger + 1, \quad \text{etc...}$$

$$= \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \left[6(n+1)(n+2) - 12(n+1) + 3\right]$$

since $|n\rangle = \frac{1}{\sqrt{n!}} a^{\dagger n}|0\rangle$

$$a^\dagger|n\rangle = \frac{1}{\sqrt{n!}} a^{\dagger\, n+1}|0\rangle = \sqrt{n+1}\,|n+1\rangle$$

$$\langle n | a a^\dagger | n \rangle = n+1$$

$$a^\dagger a^\dagger | n \rangle = \sqrt{(n+1)(n+2)}\, | n+2 \rangle$$

$$\langle n | a a a^\dagger a^\dagger | n \rangle = (n+1)(n+2)$$

Thus

$$\langle n | x^4 | n \rangle = \frac{1}{4}\left(\frac{\hbar}{m\omega}\right)^2 \left[6(n+1)(n+2-2) + 3 \right]$$

$$= \frac{3}{2}\left(\frac{\hbar}{m\omega}\right)^2 \left[n(n+1) + \frac{1}{2} \right]$$

S7.4.6 By Ehrenfest's theorem

$$\frac{d}{dt}\langle a(t)\rangle \equiv \frac{d}{dt}\langle \psi(t)|a|\psi(t)\rangle$$

$$= \frac{-i}{\hbar}\langle \psi(t)|[a, H]|\psi(t)\rangle$$

$$= \frac{-i}{\hbar}\langle \psi(t)|[a, \hbar\omega(aa^{\dagger} + \tfrac{1}{2})]|\psi(t)\rangle$$

$$= -i\omega \langle \psi(t)|a|\psi(t)\rangle = -i\omega\langle a(t)\rangle$$

Therefore

$$\langle a(t)\rangle = e^{-i\omega t}\langle a(0)\rangle$$

And

$$\langle a^{\dagger}(t)\rangle \equiv \langle \psi(t)|a^{\dagger}|\psi(t)\rangle$$

$$= \langle \psi(t)|a|\psi(t)\rangle^{*}$$

$$= \left(e^{-i\omega t}\langle a(0)\rangle\right)^{*}$$

$$= e^{i\omega t}\langle a^{\dagger}(0)\rangle .$$

S 7.4.5

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$

i. $$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-i/\hbar E_0 t}|0\rangle + e^{-i/\hbar E_1 t}|1\rangle\right)$$

$$= \frac{1}{\sqrt{2}} e^{-i\omega/2 t}\left(|0\rangle + e^{-i\omega t}|1\rangle\right)$$

ii. $$\langle X(0)\rangle = \frac{1}{2}\left(\langle 0| + \langle 1|\right)\frac{\Delta}{\sqrt{2}}(a + a^\dagger)\left(|0\rangle + |1\rangle\right)$$

$$= \frac{1}{2\sqrt{2}}\Delta\left(\langle 0|a|1\rangle + \langle 1|a^\dagger|0\rangle\right) \qquad \Delta = \left(\frac{\hbar}{m\omega}\right)^{1/2}$$

$$= \frac{1}{\sqrt{2}}\Delta$$

$$\langle P(0)\rangle = \frac{1}{2}\left(\langle 0| + \langle 1|\right)\frac{\hbar}{\sqrt{2}\Delta i}(a - a^\dagger)\left(|0\rangle + |1\rangle\right)$$

$$= \frac{\hbar}{2\sqrt{2}\Delta i}\left(\langle 0|a|1\rangle - \langle 1|a^\dagger|0\rangle\right) = 0$$

since $\quad a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$

$$\langle X(t) \rangle = \frac{1}{2\sqrt{2}} \Delta \left[\langle 0|a|1\rangle e^{-i\omega t} + \langle 1|a^\dagger|0\rangle e^{i\omega t} \right]$$

$$= \frac{1}{\sqrt{2}} \Delta \cos \omega t$$

$$\langle P(t) \rangle = \frac{\hbar}{2\sqrt{2}\Delta i} \left[\langle 0|a|1\rangle e^{-i\omega t} - \langle 1|a^\dagger|0\rangle e^{i\omega t} \right]$$

$$= \frac{-\hbar}{\sqrt{2}\Delta} \sin \omega t$$

<u>iii.</u> Ehrenfest's theorem tells us that

$$\left\{ \begin{array}{l} \dfrac{d}{dt} \langle X(t) \rangle = \dfrac{1}{m} \langle P(t) \rangle \\[2ex] \dfrac{d}{dt} \langle P(t) \rangle = - \left\langle \dfrac{dV}{dX} \right\rangle = - m\omega^2 \langle X(t) \rangle \end{array} \right.$$

Thus $$\frac{d^2}{dt^2} \langle X(t) \rangle = -\omega^2 \langle X(t) \rangle$$

Therefore, in an arbitrary state,

$$\langle x(t) \rangle \equiv \langle \psi(t) | X | \psi(t) \rangle$$

$$= A \cos(\omega t + \delta)$$

$$\langle P(t) \rangle \equiv \langle \psi(t) | P | \psi(t) \rangle$$

$$= m \frac{d}{dt} \langle x(t) \rangle = -A m \omega \sin(\omega t + \delta)$$

This agrees with $\underline{ii}$.

For $|\psi(0)\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$,

$$A = \frac{1}{\sqrt{2}} \Delta \quad \text{and} \quad \delta = 0$$

S 7.4.9

$$X \to x$$

$$P \to \frac{\hbar}{i}\frac{d}{dx} + f(x)$$

i. $$[X, P] = \left[x, \frac{\hbar}{i}\frac{d}{dx} + f(x)\right]$$

$$= \left[x, \frac{\hbar}{i}\frac{d}{dx}\right] = i\hbar$$

ii. $$|x\rangle \to |\tilde{x}\rangle = e^{ig(x)/\hbar}|x\rangle$$

with $$g(x) = \int^{x} f(x')\,dx' .$$

$$\langle\tilde{x}|X|\psi\rangle = \langle x|e^{-ig(x)/\hbar} X|\psi\rangle$$

$$= x\langle\tilde{x}|\psi\rangle$$

$$\langle \tilde{x} | P | \psi \rangle = \langle x | e^{-ig(x)/\hbar} P | \psi \rangle$$

$$= e^{-ig(x)/\hbar} \langle x | P | \psi \rangle$$

$$= e^{-ig(x)/\hbar} \frac{\hbar}{i} \frac{d}{dx} \langle x | \psi \rangle$$

$$= \left(\frac{\hbar}{i} \frac{d}{dx} + \frac{dg}{dx} \right) e^{-ig(x)/\hbar} \langle x | \psi \rangle$$

$$= \left(\frac{\hbar}{i} \frac{d}{dx} + f(x) \right) \langle \tilde{x} | \psi \rangle$$

This is true in particular when

$$|\psi\rangle = |\tilde{x}'\rangle .$$

In that case

$$\langle \tilde{x} | \tilde{x}' \rangle = e^{-i(g(x) - g(x'))/\hbar} \langle x | x' \rangle$$

$$= \delta(x - x') .$$